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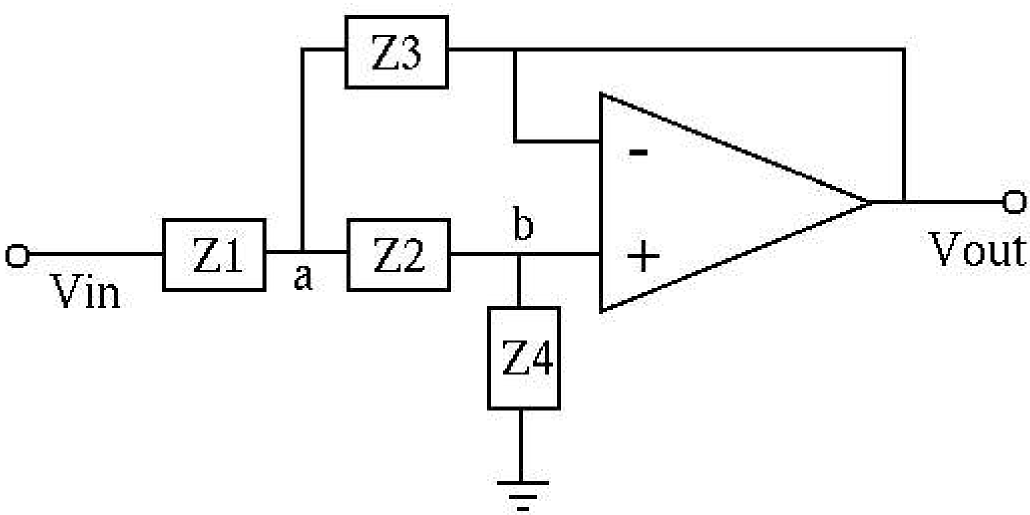
# The Sallen-Key filters

The Sallen-Key filters are second-order active filters (low-pass, high-pass, and band-pass) that can be easily implemented using the configuration below:




We represent all voltages in phasor form. Due to the virtual ground assumption, $V_b \approx V_{out}$ at non-inverting input is virtually the same as that at the inverting input, which is connected to the output $V_{out}$. Applying KCL to nodes a and b to get:

$$\frac{V_a-V_{in}}{Z_1}+\frac{V_a-V_{out}}{Z_3}+\frac{V_a-V_{out}}{Z_2}=\frac{V_a-V_{in}}{Z_1}+(V_a-V_{out})\left(\frac{1}{Z_3}+\frac{1}{Z_2}\right)=0$$

and

$$\frac{V_{out}-V_a}{Z_2}+\frac{V_{out}}{Z_4}=0,$$

The second equation can also be written as

$$V_a-V_{out}=V_{out}\frac{Z_2}{Z_4}, \qquad V_a=V_{out}\frac{Z_2+Z_4}{Z_4}$$

Substituting these into the first equation we get

$$V_{out}\ \left(\frac{Z_2+Z_4}{Z_1Z_4}+\frac{Z_2}{Z_3Z_4}+\frac{Z_2}{Z_2Z_4}\right)=\frac{V_{in}}{Z_1}$$

Now the frequency response of the Sallen-Key filter can be found as the ratio of $V_{out}$ and $V_{in}$:

$$H = \frac{V_{out}}{V_{in}}=\frac{Z_3Z_4}{Z_1Z_2+Z_1Z_3+Z_2Z_3+Z_3Z_4}$$

- **Second order low-pass filter**

  If we let $Z_1=R_1$, $Z_2=R_2$, $Z_3=1/j\omega C_1$, $Z_4=1/j\omega C_2$, then the FRF becomes:

$$H(j\omega) = \frac{1/(j\omega)^2C_1C_2}{R_1R_2+(R_1+R_2)/j\omega C_1+1/(j\omega)^2C_1C_2}$$

$$= \frac{1/R_1R_2C_1C_2}{(j\omega)^2+j\omega(R_1+R_2)/R_1R_2C_1+1/R_1R_2C_1C_2}$$

$$= \frac{1/R_1R_2C_1C_2}{(j\omega)^2+j\omega/R_pC_1+1/R_1R_2C_1C_2}$$

$$= \frac{\omega_n^2}{(j\omega)^2+2\zeta\omega_n\ j\omega+\omega_n^2}=\frac{\omega_n^2}{(j\omega)^2+\omega_n/Q\ j\omega+\omega_n^2}=\frac{\omega_n^2}{(j\omega)^2+\Delta\omega\ j\omega+\omega_n^2}$$

  This is a 2nd-order low-pass filter with

$$\omega_n=\frac{1}{\sqrt{R_1R_2C_1C_2}},\qquad \Delta\omega=\frac{1}{(R_1||R_2)C_1}=\frac{1}{R_pC_1},\qquad R_p=R_1||R_2=\frac{R_1R_2}{R_1+R_2}$$

  and

$$Q=\frac{\omega_n}{\Delta\omega}=\frac{\sqrt{R_1R_2C_1C_2}}{(R_1+R_2)C_2},\qquad \zeta=\frac{1}{2Q}=\frac{(R_1+R_2)C_2}{2\sqrt{R_1R_2C_1C_2}}$$

  As there are only two parameters $\omega_n$ and $\zeta$ or $Q=1/2\zeta$ to satisfy, we can arbitrarily set any two of the four variables $R_1$, $R_2$, $C_1$, and $C_2$, and then solve for the other two. For example, for convenience, if we let $R_1=R_2=1$, we get

$$\omega_n = \frac{1}{\sqrt{C_1 C_2}}, \qquad \Delta\omega = \frac{1}{C_1}$$

- **Second order high-pass filter**

  If we let $Z_1 = 1/j\omega C_1$, $Z_2 = 1/j\omega C_2$, $Z_3 = R_1$, $Z_4 = R_2$, then the FRF becomes becomes:

$$\begin{aligned}
H(j\omega) &= \frac{R_1 R_2}{1/(j\omega)^2 C_1 C_2 + R_1(1/j\omega C_1 + 1/j\omega C_2) + R_1 R_2} \\
&= \frac{(j\omega)^2}{1/R_1 R_2 C_1 C_2 + j\omega\ (C_1 + C_2)/C_1 C_2 R_2 + (j\omega)^2} \\
&= \frac{(j\omega)^2}{(j\omega)^2 + j\omega\ /C_s R_2 + 1/R_1 R_2 C_1 C_2} \\
&= \frac{(j\omega)^2}{(j\omega)^2 + 2\zeta\omega_n\ j\omega + \omega_n^2} = \frac{(j\omega)^2}{(j\omega)^2 + \omega_n/Q\ j\omega + \omega_n^2} = \frac{(j\omega)^2}{(j\omega)^2 + \Delta\omega\ j\omega + \omega_n^2}
\end{aligned}$$

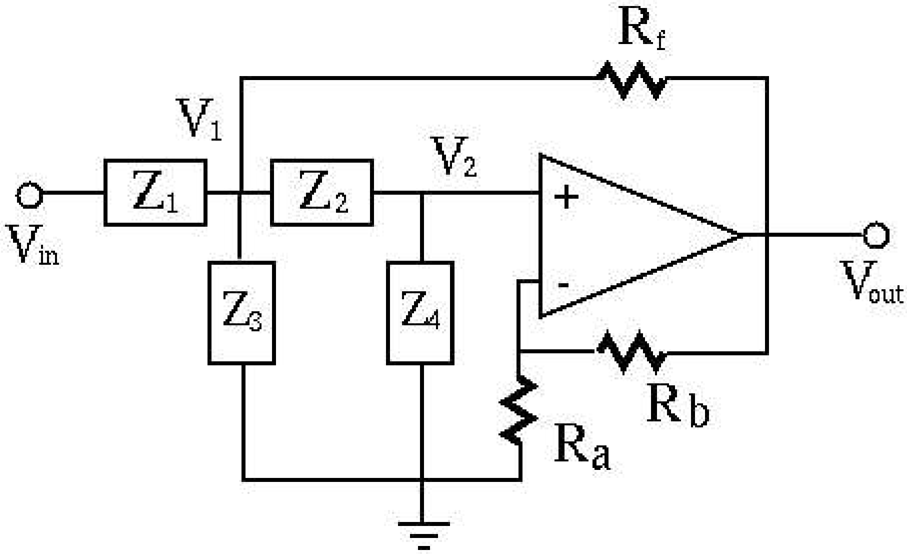
  This is a 2nd-order high-pass filter with

$$\omega_n = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \qquad \Delta\omega = \frac{1}{C_s R_2}, \qquad C_s = \frac{C_1 C_2}{C_1 + C_2}$$

  and

$$Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{(C_1 + C_2) R_1}, \qquad \zeta = \frac{1}{2Q} = \frac{(C_1 + C_2) R_1}{2\sqrt{R_1 R_2 C_1 C_2}}$$

We further consider a band-pass filter shown below:




By voltage divider and virtual ground, we get

$$V_2 = \frac{R_a}{R_a + R_b} V_{out} = k V_{out}, \qquad \left( k = \frac{R_a}{R_a + R_b} \right)$$

Apply KCL to node $V_2$ to get:

$$\frac{V_1 - V_2}{Z_2} = \frac{V_2}{Z_4}, \qquad \text{i.e.} \quad V_1 = V_2 \left( \frac{1}{Z_2} + \frac{1}{Z_4} \right) Z_2 = V_2 \left( 1 + \frac{Z_2}{Z_4} \right)$$

Apply KCL to node $V_1$ to get:

$$\frac{V_{in} - V_1}{Z_1} + \frac{V_{out} - V_1}{R_f} + \frac{V_2 - V_1}{Z_2} = \frac{V_1}{Z_3}$$

Rearrange the terms, and replace $V_1$ by $V_2(1 + Z_2/Z_4)$ to get

$$\frac{V_{in}}{Z_1} + \frac{V_{out}}{R_f} + \frac{V_2}{Z_2} = V_1 \left( \frac{1}{Z_1} + \frac{1}{R_f} + \frac{1}{Z_2} + \frac{1}{Z_3} \right)$$

$$= V_2 \left( 1 + \frac{Z_2}{Z_4} \right) \left( \frac{1}{Z_1} + \frac{1}{R_f} + \frac{1}{Z_2} + \frac{1}{Z_3} \right)$$

Further rearrange the terms and replace $V_2$ by $kV_{out}$ to get

$$\frac{V_{in}}{Z_1}+\frac{V_{out}}{R_f}=kV_{out}\left[\left(1+\frac{Z_2}{Z_4}\right)\left(\frac{1}{Z_1}+\frac{1}{R_f}+\frac{1}{Z_2}+\frac{1}{Z_3}\right)-\frac{1}{Z_2}\right]$$

Further rearrange the terms

$$\begin{aligned}\frac{V_{in}}{Z_1} &= kV_{out}\left[\left(1+\frac{Z_2}{Z_4}\right)\left(\frac{1}{Z_1}+\frac{1}{R_f}+\frac{1}{Z_2}+\frac{1}{Z_3}\right)-\frac{1}{Z_2}\right]-\frac{V_{out}}{R_f} \\ &= V_{out}\left\{k\left[\left(1+\frac{Z_2}{Z_4}\right)\left(\frac{1}{Z_1}+\frac{1}{R_f}+\frac{1}{Z_2}+\frac{1}{Z_3}\right)-\frac{1}{Z_2}\right]-\frac{1}{R_f}\right\}\end{aligned}$$

Finally we get the frequency response function:

$$\begin{aligned}H=\frac{V_{out}}{V_{in}} &= \frac{1}{Z_1\left\{k\left[\left(1+\frac{Z_2}{Z_4}\right)\left(\frac{1}{Z_1}+\frac{1}{R_f}+\frac{1}{Z_2}+\frac{1}{Z_3}\right)-\frac{1}{Z_2}\right]-\frac{1}{R_f}\right\}} \\ &= \frac{1/k}{Z_1\left[\left(1+\frac{Z_2}{Z_4}\right)\left(\frac{1}{Z_1}+\frac{1}{R_f}+\frac{1}{Z_2}+\frac{1}{Z_3}\right)-\frac{1}{Z_2}\right]-\frac{Z_1}{kR_f}} \\ &= \frac{1/k}{1+\frac{Z_1}{R_f}+\frac{Z_1}{Z_3}+\frac{Z_2}{Z_4}+\frac{Z_1Z_2}{Z_4R_f}+\frac{Z_1}{Z_4}+\frac{Z_1Z_2}{Z_3Z_4}-\frac{Z_1}{kR_f}}\end{aligned}$$

If we let

$$Z_1=R_1,\quad Z_4=R_2,\quad Z_2=\frac{1}{j\omega C_2},\quad Z_3=\frac{1}{j\omega C_1}$$

the frequency response function becomes

$$\begin{aligned}H &= \frac{\left(1+\frac{R_b}{R_a}\right)\frac{1}{R_1C_1}j\omega}{(j\omega)^2+\left(\frac{1}{R_2C_1}+\frac{1}{R_2C_2}+\frac{1}{R_1C_1}-\frac{R_b}{R_aR_fC_1}\right)j\omega+\frac{R_1+R_f}{R_fR_1R_2C_1C_2}} \\ &= \frac{Aj\omega}{(j\omega)^2+\Delta\omega j\omega+\omega_n^2},\qquad A=(1+R_b/R_a)/R_1C_1\end{aligned}$$

This is a band-pass filter with the peak frequency equal to the natural frequency:

$$\omega_n=\sqrt{\frac{R_1+R_f}{R_fR_1R_2C_1C_2}}$$

the bandwidth

$$\Delta\omega = \frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_1 C_1} - \frac{R_b}{R_a R_f C_1}$$

The gain of the filter is controlled by $1 + R_b/R_a$.

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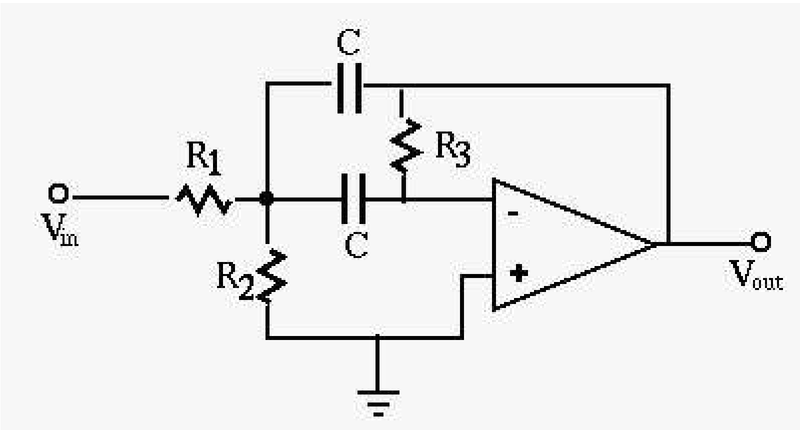
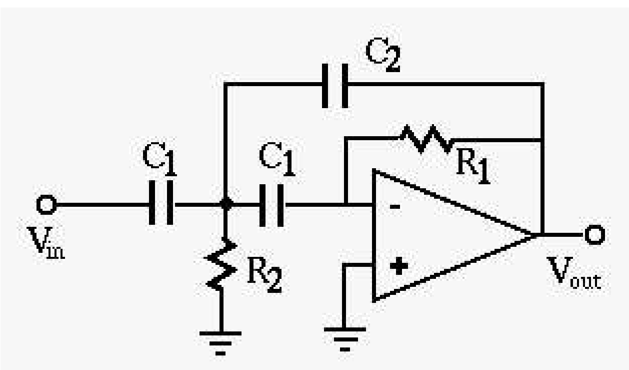
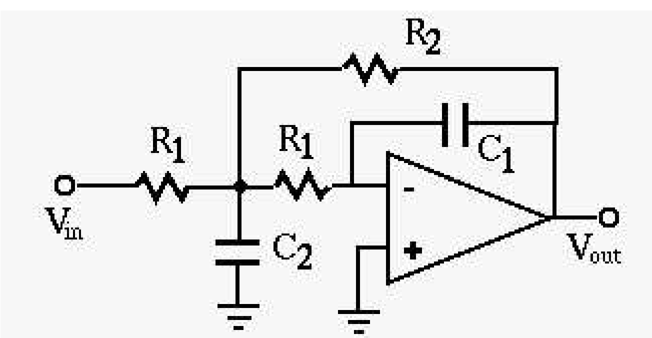
More Examples of op-amp filters are listed below (with detailed analysis in here):

- 

- 

- 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*Ruye Wang 2019-05-07*